



Database:- A database is a collection of data organized of easy access and management.

Types of Database:-

1. Relational Database -----> SQL
2. Non Relational Database -----> NoSQL

Difference of Relational and Non Relational Database:-

- Relational DB is stores data in “Structure way”(Raw & Coloum).
- Non Relational DB is stores data in “No Structure way”(Key:value).

Tools

SQL	NoSQL
MySQL	Mongo DB
MariaDB	Raven DB
PostgrSQL	Dynamo DB
Oracle DB	Apache (assendra)
Auror DB	Neo4J
IBM DB	

MariaDB Database Server :-

- Database installation and configuration.
- Secure Installation.
- Database creation & Management.
- DB Permission.
- DB backup & Restore
- Basic DB Troubleshooting commands.

Setup MariaDB

- Package name ---> mariadb-server
- Service ---> mariadb
- Database store ---> /var/lib/mysql
- Configure file ----> /etc/my.cnf
- Port ----> 3306

Commands

- 1 create database manish; --> to create a database (manish – DB name)
- 2 show databases; --> show all databases
- 3 drop database manish; --> to delete a database
- 4 use manish; --> come inside database
- 5 show tables; --> to list all tables inside the manish database
- 6 describe user; --> to describe user's tables
- 7 grant --> to give permission to user
- 8 revoke --> to remove permissions
- 9 select user from user; --> to check the user table into user filed
- 10 select user,password,host from user; --> list fileds

Install Mariadb

```
[root@Database-server ~]# yum install mariadb-server -y
```

```
[root@Database-server ~]# systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
   Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-09-21 02:59:55 UTC; 13s ago
     Docs: man:mariadb(8)
           https://mariadb.com/kb/en/library/systemd/
   Process: 17528 ExecStartPre=/usr/libexec/mariadb-check-socket (code=exited, status=0/SUCCESS)
   Process: 17550 ExecStartPre=/usr/libexec/mariadb-prepare-db-dir mariadb.service (code=exited, status=0/SUCCESS)
   Process: 17644 ExecStartPost=/usr/libexec/mariadb-check-upgrade (code=exited, status=0/SUCCESS)
   Main PID: 17632 (mariabdd)
   Status: "Taking your SQL requests now..."

[root@Database-server ~]# mysql --version
mysql Ver 15.1 Distrib 10.5.27-MariaDB, for Linux (x86_64) using EditLine wrapper
[root@Database-server ~]#
```

`mysql -u root (root for database server )`

```
[root@Database-server ~]# mysql -u root
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

`show databases;`

```
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
+-----+
3 rows in set (0.001 sec)

MariaDB [(none)]>
```

`create database manish;`

```

MariaDB [(none)]> create database maish;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| maish      |
| mysql      |
| performance_schema |
+-----+
4 rows in set (0.000 sec)

MariaDB [(none)]>

```

drop database manish;

```

MariaDB [(none)]> drop database maish;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql      |
| performance_schema |
+-----+
3 rows in set (0.000 sec)

MariaDB [(none)]>

```

use mysql;

```

MariaDB [(none)]> use mysql;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [mysql]> S_

```

show tables;

```
MariaDB [mysql]> Show tables;
+-----+
| Tables_in_mysql |
+-----+
| column_stats    |
| columns_priv    |
| db              |
| event           |
| func            |
| general_log     |
| global_priv     |
| gtid_slave_pos  |
| help_category   |
| help_keyword    |
| help_relation   |
| help_topic      |
| index_stats     |
| innodb_index_stats |
| innodb_table_stats |
| plugin          |
| proc            |
| procs_priv      |
| proxies_priv    |
| roles_mapping   |
| servers         |
| slow_log        |
| table_stats     |
| tables_priv     |
| time_zone       |
| time_zone_leap_second |
| time_zone_name  |
| time_zone_transition |
| time_zone_transition_type |
| transaction_registry |
| user            |
+-----+
31 rows in set (0.000 sec)

MariaDB [mysql]>
```

describe user;

ERROR 1146 (42S02): Table 'mysql.users' doesn't exist

MariaDB [mysql]> describe user;

Field	Type	Null	Key	Default	Extra
Host	char(60)	NO			
User	char(80)	NO			
Password	longtext	YES		NULL	
Select_priv	varchar(1)	YES		NULL	
Insert_priv	varchar(1)	YES		NULL	
Update_priv	varchar(1)	YES		NULL	
Delete_priv	varchar(1)	YES		NULL	
Create_priv	varchar(1)	YES		NULL	
Drop_priv	varchar(1)	YES		NULL	
Reload_priv	varchar(1)	YES		NULL	
Shutdown_priv	varchar(1)	YES		NULL	
Process_priv	varchar(1)	YES		NULL	
File_priv	varchar(1)	YES		NULL	
Grant_priv	varchar(1)	YES		NULL	
References_priv	varchar(1)	YES		NULL	
Index_priv	varchar(1)	YES		NULL	
Alter_priv	varchar(1)	YES		NULL	
Show_db_priv	varchar(1)	YES		NULL	
Super_priv	varchar(1)	YES		NULL	
Create_tmp_table_priv	varchar(1)	YES		NULL	
Lock_tables_priv	varchar(1)	YES		NULL	
Execute_priv	varchar(1)	YES		NULL	
Repl_slave_priv	varchar(1)	YES		NULL	
Repl_client_priv	varchar(1)	YES		NULL	
Create_view_priv	varchar(1)	YES		NULL	
Show_view_priv	varchar(1)	YES		NULL	
Create_routine_priv	varchar(1)	YES		NULL	
Alter_routine_priv	varchar(1)	YES		NULL	

select user from user;

MariaDB [mysql]> select user from user;

User
mariadb.sys
mysql
root

3 rows in set (0.001 sec)

MariaDB [mysql]>

Secure Installation for DB:-

Secure installation in DB means configuring the database safely so that only authorized users can access it and data stays protected.

```
[root@Database-server ~]#
[root@Database-server ~]# mysql_secure_installation

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!

In order to log into MariaDB to secure it, we'll need the current
password for the root user. If you've just installed MariaDB, and
haven't set the root password yet, you should just press enter here.

Enter current password for root (enter for none):
OK, successfully used password, moving on...

Setting the root password or using the unix_socket ensures that nobody
can log into the MariaDB root user without the proper authorisation.

You already have your root account protected, so you can safely answer 'n'.

Switch to unix_socket authentication [Y/n] y
Enabled successfully!
Reloading privilege tables..
... Success!

You already have your root account protected, so you can safely answer 'n'.

Change the root password? [Y/n] y
New password:
Re-enter new password:
Password updated successfully!
Reloading privilege tables..
... Success!

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them. This is intended only for testing, and to make the installation
go a bit smoother. You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] y
... Success!
```


By default, a MariaDB installation has an anonymous user, allowing anyone to log into MariaDB without having to have a user account created for them. This is intended only for testing, and to make the installation go a bit smoother. You should remove them before moving into a production environment.

```
Remove anonymous users? [Y/n] y
... Success!
```

Normally, root should only be allowed to connect from 'localhost'. This ensures that someone cannot guess at the root password from the network.

```
Disallow root login remotely? [Y/n] y
... Success!
```

By default, MariaDB comes with a database named 'test' that anyone can access. This is also intended only for testing, and should be removed before moving into a production environment.

```
Remove test database and access to it? [Y/n] y
- Dropping test database...
... Success!
- Removing privileges on test database...
... Success!
```

Reloading the privilege tables will ensure that all changes made so far will take effect immediately.

```
Reload privilege tables now? [Y/n] y
... Success!
```

Cleaning up...

All done! If you've completed all of the above steps, your MariaDB installation should now be secure.

```
Thanks for using MariaDB!
[root@Database-server ~]#
```

After secure installation run the command

mysql -u root -p

```
[root@Database-server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 14
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

User Management:-

User Management in DB is the process of handling user accounts, their authentication, authorization, and privileges.

- ★ @localhost – this user can never login remotely.
- ★ @% --> this user login any network.
- ★ @IP address --> this user login in a particular IP.

To create users in database server:-

- 1 create user manish@localhost identified by 'password';
- 2 create user kunal@'%' identified by 'password';
- 3 create user sunil@'173.34.0.0' identified by 'password';

```
MariaDB [(none)]> create user mansih@localhost identified by 'password';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> create user kunal@'%' identified by 'password';
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]> create user sunil@'173.34.0.0' identified by 'password';
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> _
```

Show the users :-

use mysql --> show tables ---> select user,password,host from user;

```
MariaDB [(none)]> use mysql;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
MariaDB [mysql]> show tables;
```

Tables_in_mysql
column_stats
columns_priv
db
event
func
general_log
global_priv
gtid_slave_pos
help_category
help_keyword
help_relation
help_topic
index_stats
innodb_index_stats
innodb_table_stats
plugin
proc
procs_priv
proxies_priv
roles_mapping
servers
slow_log
table_stats
tables_priv
time_zone
time_zone_leap_second
time_zone_name
time_zone_transition
time_zone_transition_type
transaction_registry
user

```
31 rows in set (0.000 sec)
```

```

MariaDB [mysql]> select user,password,host from user;
+-----+-----+-----+
| User      | Password                                     | Host      |
+-----+-----+-----+
| mariadb.sys | *6260AD0114EA0D40D08B8726D34E07DB6AE4B428 | localhost |
| root       | *6260AD0114EA0D40D08B8726D34E07DB6AE4B428 | localhost |
| mysql      | invalid                                     | localhost |
| mansih     | *2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19 | localhost |
| kunal      | *2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19 | %         |
| sunil      | *2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19 | 173.34.0.0 |
+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [mysql]>

```

Give the asses particular database to particular user:-

Create a database

```

MariaDB [(none)]> create database ecommerce;
Query OK, 1 row affected (0.000 sec)

```

Grant all on ecommerce.* to mansih@localhost;

```

MariaDB [(none)]> grant all on ecommerce.* to mansih@localhost;
Query OK, 0 rows affected (0.002 sec)

MariaDB [(none)]> show grants for mansih@localhost;
+-----+-----+-----+
| Grants for mansih@localhost |
+-----+-----+-----+
| GRANT USAGE ON *.* TO `mansih`@`localhost` IDENTIFIED BY PASSWORD '*2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19' |
| GRANT ALL PRIVILEGES ON `ecommerce`.* TO `mansih`@`localhost` |
+-----+-----+-----+
2 rows in set (0.000 sec)

MariaDB [(none)]>

```

Remove permission on Dtabase

Revoke all on ecommerce.* from mansih@localhost;

```

MariaDB [(none)]> revoke all on ecommerce.* from mansih@localhost;
Query OK, 0 rows affected (0.001 sec)

```

Check it

Show grants from mansih@localhost;

```
MariaDB [(none)]> show grants for mansih@localhost;
+-----+
| Grants for mansih@localhost |
+-----+
| GRANT USAGE ON *.* TO `mansih`@`localhost` IDENTIFIED BY PASSWORD '*2470C0C06DEE42FD1618BB99005ADCA2EC9D1E19' |
+-----+
1 row in set (0.000 sec)

MariaDB [(none)]> _
```

To create tables in Dtabase:-

Use fintaetch;

Create table students(name varchar(50),course varchar(50),phone int(20));

```
MariaDB [fintach]> create table student(name varchar(50),course varchar(50),phone int(20));
Query OK, 0 rows affected (0.007 sec)

MariaDB [fintach]>
```

Show tables

```
MariaDB [fintach]> show tables;
+-----+
| Tables_in_fintach |
+-----+
| student            |
+-----+
1 row in set (0.000 sec)

MariaDB [fintach]> S_
```

Describe student;

```
MariaDB [fintach]> describe student;
```

Field	Type	Null	Key	Default	Extra
name	varchar(50)	YES		NULL	
course	varchar(50)	YES		NULL	
phone	int(20)	YES		NULL	

```
3 rows in set (0.001 sec)
```

```
MariaDB [fintach]> _
```

Insert into student values('manish','Devopss','986958698');

```
MariaDB [fintach]> insert into student values('manish','DevOps','99800989');  
Query OK, 1 row affected (0.001 sec)
```

```
MariaDB [fintach]> _
```

Check table

Select name,course,phone from student;

```
MariaDB [fintach]> select name,course,phone from student;
```

name	course	phone
manish	DevOps	99800989

```
1 row in set (0.001 sec)
```

```
MariaDB [fintach]> _
```

```
MariaDB [fintach]> insert into student values('kunal','Cloud','99800989');
Query OK, 1 row affected (0.001 sec)

MariaDB [fintach]> insert into student values('sunil','Python','99800989');
Query OK, 1 row affected (0.001 sec)

MariaDB [fintach]> select name,course,phone from student;
+-----+-----+-----+
| name  | course | phone  |
+-----+-----+-----+
| manish | DevOps | 99800989 |
| kunal  | Cloud  | 99800989 |
| sunil  | Python | 99800989 |
+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [fintach]>
```

Taking the backup for database:-


```

[root@Database-server ~]# mysqldump fintach > ecommerce.sql
[root@Database-server ~]# mysql -u -root -p
Enter password:
ERROR 1045 (28000): Access denied for user '-root'@'localhost' (using password: YES)
[root@Database-server ~]# mysql -u -root -p
Enter password:
ERROR 1045 (28000): Access denied for user '-root'@'localhost' (using password: YES)
[root@Database-server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 22
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> drop database fintach;
Query OK, 1 row affected (0.004 sec)

MariaDB [(none)]> show database;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near 't line 1' at line 1
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| ecommerce |
| information_schema |
| mysql |
| performance_schema |
+-----+

```

Restore the Backup:-

```
MariaDB [(none)]> ^DBye
[root@Database-server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 23
Server version: 10.5.27-MariaDB MariaDB Server

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database backup;
Query OK, 1 row affected (0.000 sec)

MariaDB [(none)]> ^DBye
[root@Database-server ~]# mysql backup < ecommerce.sql
-bash: ecommerce.sql: No such file or directory
[root@Database-server ~]# ls
ecommerce.sql
[root@Database-server ~]# mysql backup < ecommerce.sql
[root@Database-server ~]#
```

```
[root@Database-server ~]# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 25
Server version: 10.5.27-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| backup   |
| ecommerce |
| information_schema |
| mysql    |
| performance_schema |
+-----+
5 rows in set (0.000 sec)

MariaDB [(none)]> S
```

```
MariaDB [(none)]> use backup;  
Reading table information for completion of table and column names  
You can turn off this feature to get a quicker startup with -A  
  
Database changed  
MariaDB [backup]> show tables;  
+-----+  
| Tables_in_backup |  
+-----+  
| student          |  
+-----+  
1 row in set (0.000 sec)  
  
MariaDB [backup]>
```

```
MariaDB [backup]> select name,course,phone from student;  
+-----+-----+-----+  
| name  | course | phone  |  
+-----+-----+-----+  
| manish | DevOps | 99800989 |  
| kunal  | Cloud  | 99800989 |  
| sunil  | Python | 99800989 |  
+-----+-----+-----+  
3 rows in set (0.000 sec)
```

Complete Database

Thank you 🧐

Happy learning



By Manish Saini